

Scenario:

Karen is a security professional looking for a new job. A company called "TAAUSA" offered her a position and asked her to complete a couple of tasks to prove her technical competency. As a soc analyst Analyze the provided disk image and answer the questions based on your understanding of the cases she was assigned to investigate.

CTF link : <https://cyberdefenders.org/blueteam-ctf-challenges/hireme/>

Tools used : FTK Imager,Registry Explorer,OST Viewer

Q1-What is the administrator's username?

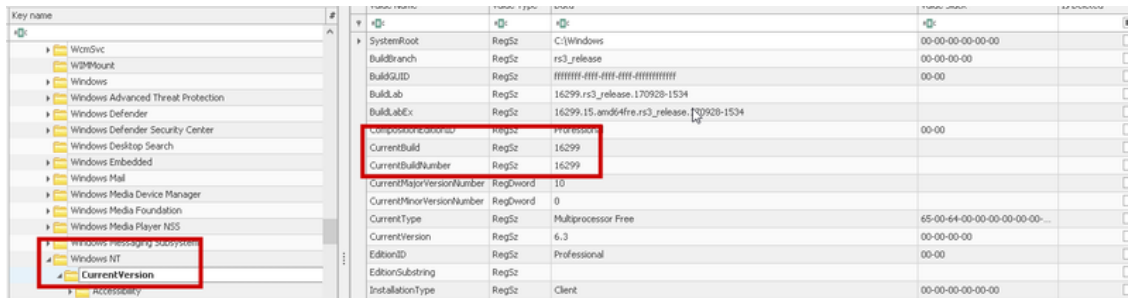
using FTK Imager to add a .ad1 image we first click on Add New Evidence Item then we Choose the Image File then add the path of our image after loading the image we explore the Evidence Tree and easily get the username



so the answer here is :Karen

Q2-What is the OS's build number?

using registry explorer to get the OS build number we need to extract the SYSTEM hive from FTK imager we can locate it under windows/system32/config and then look under the path SOFTWARE/Microsoft/WindowsNT/CurrentVersion

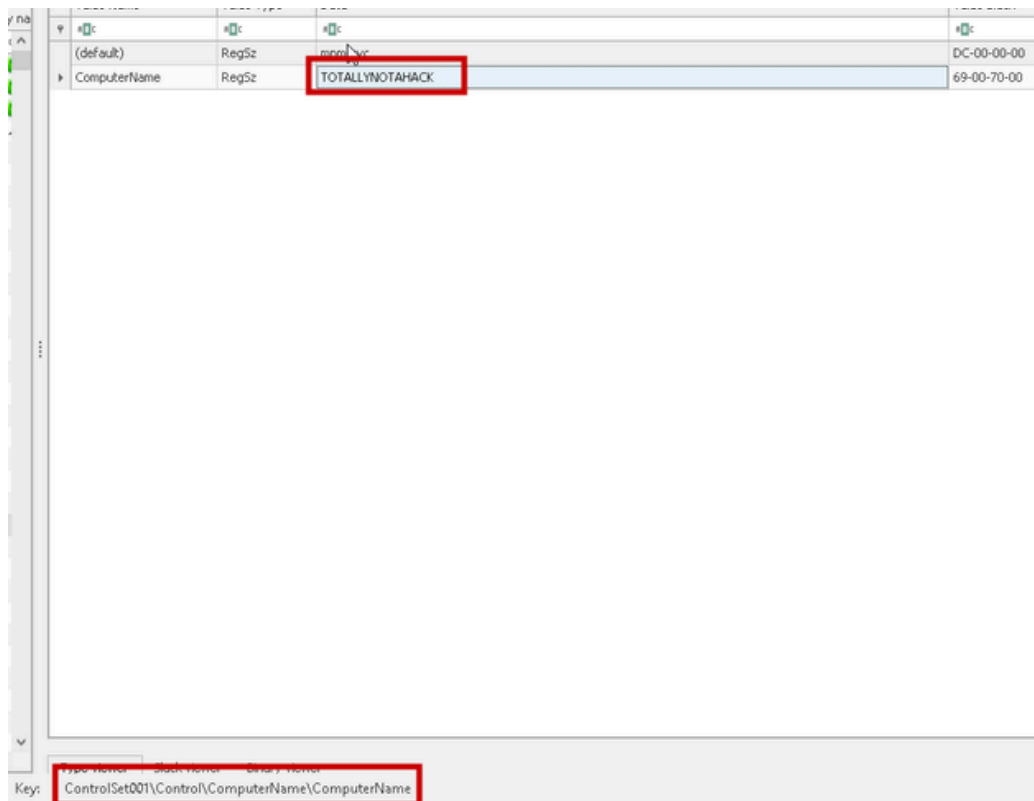


Key name	Value name	Value type	Value	Value data	Value type
SystemRoot	RegSz	C:\Windows		00-00-00-00-00-00	
BuildBranch	RegSz	rs3_release		00-00-00-00	
BuildGUID	RegSz	{f1f1f1f1-f1f1-f1f1-f1f1-f1f1f1f1}		00-00	
BuildLab	RegSz	16299.rs3_release.170920-1534			
BuildLabEx	RegSz	16299.15.amd64fre.rs3_release.170920-1534			
CompositionMode	RegSz	Professional		00-00	
CurrentBuild	RegSz	16299			
CurrentBuildNumber	RegSz	16299			
CurrentMajorVersionNumber	RegDword	10			
CurrentMinorVersionNumber	RegDword	0			
CurrentType	RegSz	Multiprocessor Free		65-00-64-00-00-00-00-00-...	
CurrentVersion	RegSz	6.3		00-00-00-00	
EditionID	RegSz	Professional		00-00	
EditionSubString	RegSz				
InstallationType	RegSz	Client		00-00-00-00-00-00	

so the answer here is : 16299

Q3-What is the hostname of the computer?

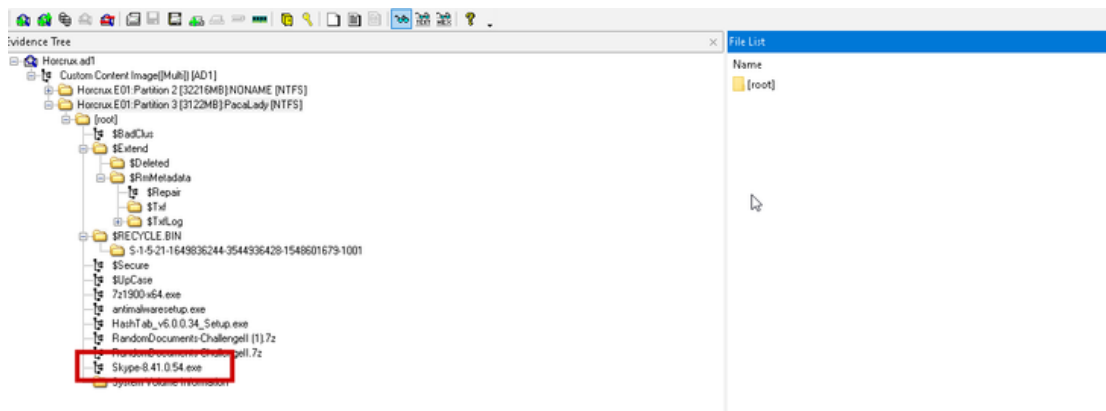
the hostname is located in the SYSTEM hive exactly under the path
SYSTEM/ControlSet001/Control/ComputerName/ComputerName



so the answer here is :TOTALLYNOTAHACK

Q4-A messaging application was used to communicate with a fellow Alpaca enthusiast. What is the name of the software?

2 partitions were mounted when we added the image if we looked inside the other partition we will find the software



so the answer here is : Skype

Q5-What is the zip code of the administrator's post?

a zip code could indicate billing information and after looking in the evidence tree we found that the victim was using chrome so to find billing information we look inside the path Karen/AppData/Local/Google/Chrome/UserData/Default as it has all the possible data about the user using chrome and if we want to get information about the user data we can look inside either Login Data file or Web Data file in our case we found the data we need in Web Data file but to get the information we need to use an SQL viewer because these files are SQLite databases after we view it we find that the victim been using auto fill to add their data

The screenshot shows the AccessData FTK Imager 4.7.1.2 interface. The 'Evidence Tree' on the left shows the file structure, with 'Web Data' highlighted. The 'File List' on the right shows various files, with 'Web Data' and 'autofill' highlighted. The 'Web Data' table is selected in the SQL viewer, showing 5 records. The 'postal' column in the third record is highlighted, showing the value '19709'.

	name	value	value_lower	date_created	date_last_used	count
1	email	klavespizza@outloo...	klavespizza@outloo...	1549740331	1549740331	1
2	PostingTitle	Job Needed. 19709	job needed. 19709	1549740599	1549740599	1
3	postal	19709	19709	1549740599	1549740599	1
4	FromEMail	klavespizza@outloo...	klavespizza@outloo...	1549740599	1549740599	1
5	ConfirmEMail	klavespizza@outloo...	klavespizza@outloo...	1549740599	1549740599	1

so the answer here is : 19709

Q6-What are the initials of the person who contacted the admin user from TAAUSAI?

looking in the evidence tree we found that the victim was using outlook looking inside the outlook folder we find a .ost file (offline storage table) which acts as a local copy of ur mailbox using OST Viewer we can get the inbox of the victim and sort it by date then we look for TAAUSAI we find someone called Micheal Scotch and in other emails sent by Karen we see it abbreviated to MS

The screenshot displays an Outlook interface. The left sidebar shows the folder structure, including 'Inbox'. The main pane shows a list of emails. One email is highlighted with a red box, showing details for a 'Job Offer High paying' from 'Alpaca Activists'. Below this, another email is shown with a red box around the name 'Best, Micheal Scotch'. The bottom pane shows a detailed view of an email from 'Karen Alice' with a red box around the text 'MS'.

From	Subject	Date/Time	Lost/Deleted
Dropbox<no-reply@dropbox.com>	Download the Dropbox desktop app	Wed 03/13/2019 20:44 PM	Existing
Dropbox<no-reply@dropbox.com>	Complete your Dropbox setup	Sun 03/10/2019 20:44 PM	Existing
Alpaca Activists<706647539f@30539e2e43ba5fd21606@reply.craigslit.org>	Job Offer High paying	Fri 03/08/2019 05:16 AM	Existing
Microsoft<1111c28029338c33604c52...>	Job Needed, 19709	Mon 02/25/2019 22:23 PM	Existing
Vanessa Stone<af46166390be3dbef4536...>	Serious Employment Opportunity From...	Mon 02/25/2019 16:10 PM	Existing
craigslit reply 5d8e5d8e1562c3a302ca...	Job Needed, 19709	Tue 02/19/2019 09:16 AM	Lost/Deleted
Jeff Astrologer<deea7ab4622302da2c14ba...>	Job Needed, 19709	Wed 02/13/2019 15:27 PM	Lost/Deleted
Dropbox<no-reply@dropbox.com>	Almost there! Only 3 steps left to Dropb...	Wed 02/13/2019 14:31 PM	Lost/Deleted
E d Kibum<ef3794f3b04b304a0273632bc...>	Job Needed, 19709	Mon 02/11/2019 18:15 PM	Lost/Deleted
Dropbox<no-reply@dropbox.com>	Which Dropbox plan is right for you?	Mon 02/11/2019 17:46 PM	Lost/Deleted
Microsoft Store<microsoft@microsoft.com>	Congrat! Your Microsoft account is w...	Mon 02/11/2019 23:26 PM	Lost/Deleted
Brent Post<ear96bdf37e2947b1572300d...>	Job Needed, 19709	Sun 02/10/2019 23:26 PM	Lost/Deleted
Thomas Miller<ec5e690296d3d4d9a025a...>	Job Needed, 19709	Sun 02/10/2019 20:16 PM	Lost/Deleted
Jason Stewart<34b1c051374c34a3989b1b...>	Job Needed, 19709	Sun 02/10/2019 00:10 AM	Lost/Deleted
craigslit - automated message, do not reply...	POST/EDIT/DELETE Job Needed, 1...	Sat 02/09/2019 23:12 PM	Lost/Deleted

Job Offer High paying
Alpaca Activists <706647539f@30539e2e43ba5fd21606@reply.craigslit.org>
To: 706647539f@30539e2e43ba5fd21606@res.craigslit.org

We came across your Craigslist entry [here](#) and wanted to know if you'd be interested in a high paying technical job involving the use of computers. Don't worry about your ski with what you need.

Let us know what you think.

Best,
- Micheal Scotch

From	Subject	Date/Time	Lost/Deleted
Karen Alice<klovespizza@outlook.com>	RE: Interested in the job	Sat 03/23/2019 02:37 AM	Existing
Karen Alice<klovespizza@outlook.com>	RE: Interested in the job	Sat 03/23/2019 02:37 AM	Existing
Alpaca Activists<taausai@gmail.com>	Re: Interested in the job	Fri 03/22/2019 04:55 AM	Existing
Alpaca Activists<taausai@gmail.com>	Re: Interested in the job	Fri 03/22/2019 04:47 AM	Existing
Jashua Tetault<9f9e860d8336488b2834...>	Re: I saw your add	Sun 03/17/2019 14:35 PM	Existing
jeff astrologer<deea7ab4622302da2c14ba...>	Re: Job Needed, 19709	Sun 03/17/2019 14:20 PM	Existing
Alpaca Activists<taausai@gmail.com>	Re: Interested in the job	Sun 03/17/2019 09:44 AM	Existing
Alpaca Activists<taausai@gmail.com>	Re: Interested in the job	Sun 03/17/2019 09:19 AM	Existing
Microsoft account team<account-security-n...>	Microsoft account security info verificat...	Sun 03/17/2019 09:02 AM	Existing
System Administrator	Undeliverable: Interested in the job	Sun 03/17/2019 09:00 AM	Existing
Outlook.com Team<member_services@outlo...	Please sign in to your Outlook.com acc...	Sun 03/17/2019 09:00 AM	Existing
System Administrator	Undeliverable: Follow Up Email	Sun 03/17/2019 08:57 AM	Existing
Outlook.com Team<member_services@outlo...	Please sign in to your Outlook.com acc...	Sun 03/17/2019 08:57 AM	Existing
Alpaca Activists<13919c46dfe3e3bc0e41...>	Follow Up Email	Sun 03/17/2019 08:32 AM	Existing
Dropbox<no-reply@dropbox.com>	Organize your files with Dropbox folders	Sat 03/16/2019 20:44 PM	Existing

RE: Interested in the job
Karen Alice <klovespizza@outlook.com>
To: "Karen Alice" <klovespizza@outlook.com>
Attachments: Skype Convo.zip

MS

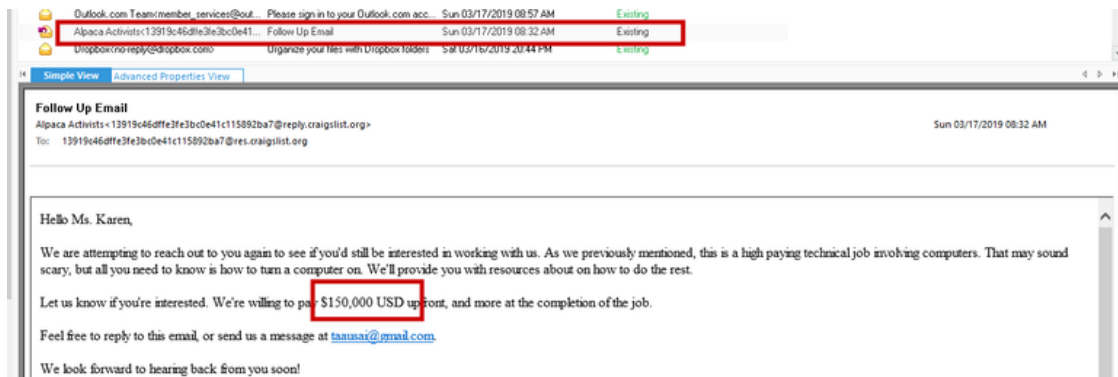
Things didn't go as planned with Bob. I attached a copy of our chat history. The password is pacalove

From: Karen Alice
Sent: 22 March 2019 01:53
To: 'Alpaca Activists' <taausai@gmail.com>
Subject: RE: Interested in the job

so the answer here is : MS

Q7-How much money was TAAUSAI willing to pay upfront?

looking through the inbox we find a Follow up email stating the money that they want to pay upfront



so the answer here is : 150000

Q8-What country is the admin user meeting the hacker group in?

looking through the inbox we find a replay email stating the location coordinates. looking up the coordinates we can find the actual location for the meetup

The screenshot shows an Outlook inbox with several emails. The first email is from Karen Alice <klovespizza@outlook.com> with the subject 'RE: Interested in the job' and the date 'Sat 03/23/2019 02:37 AM'. The email is marked as 'Lost/Deleted'. Below the inbox, the email content is displayed. It starts with 'From: Alpaca Activists <lausa@gmail.com>' and 'Sent: 22 March 2019 01:47'. The subject is 'Re: Interested in the job'. The body of the email says: 'Hey there! So here's what we need you to do: We have been conducting an investigation on Bob Redliubeht (the CEO of Alpacamybags Luxury Alpaca handbags) and we believe he's been mistreating some of his Alpacas. We have heard complaints that he refuses to provide Alpacas with scarfs and beanies during the winter! What we need you to do is gain his trust and then hack his machine. We will give you more information about this in person. Meet us her' followed by the coordinates '27°22'50.10"N, 33°37'54.62"E' in a red box. Below the email content, a map of Egypt is shown with a red pin at the location '27°22'50.1"N 33°37'54.6"E'. The map also shows the Red Sea, the Nile River, and the border with Israel. The word 'Egypt' is written in a red box on the map.

so the answer here is : Egypt

Q9-What is the machine's timezone? (Use the three-letter abbreviation)

back to registry explorer under the hive SYSTEM/ControlSet001/Control/TimeZoneInformation we can locate the timezone

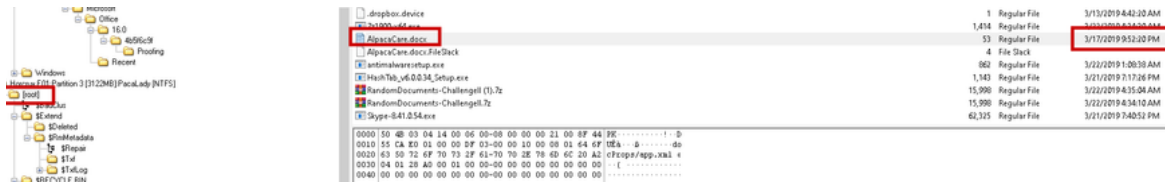
The screenshot displays the Windows Registry Editor. The left pane shows the tree structure with the path `SYSTEM\ControlSet001\Control\TimeZoneInformation` selected and highlighted with a red box. The right pane shows the details for the `Time Zone Information` key, with the `Time Zone Key Name` value highlighted and showing the value `UTC`, also highlighted with a red box. Other values like `Bias`, `DaylightBias`, `DaylightName`, `DaylightStart`, `StandardBias`, `StandardName`, `StandardStart`, `Time Zone Key Name`, and `ActiveTimeBias` are also visible.

Value Name	Value Data
Bias	0
DaylightBias	0
DaylightName	@tzres.dll,-931
DaylightStart	Month 0, week of month 0, day of week 0, Hours:Minutes:Seconds 0:0:0
StandardBias	0
StandardName	@tzres.dll,-932
StandardStart	Month 0, week of month 0, day of week 0, Hours:Minutes:Seconds 0:0:0
Time Zone Key Name	UTC
ActiveTimeBias	0

so the answer here is : UTC

Q10-When was AlpacaCare.docx last accessed?

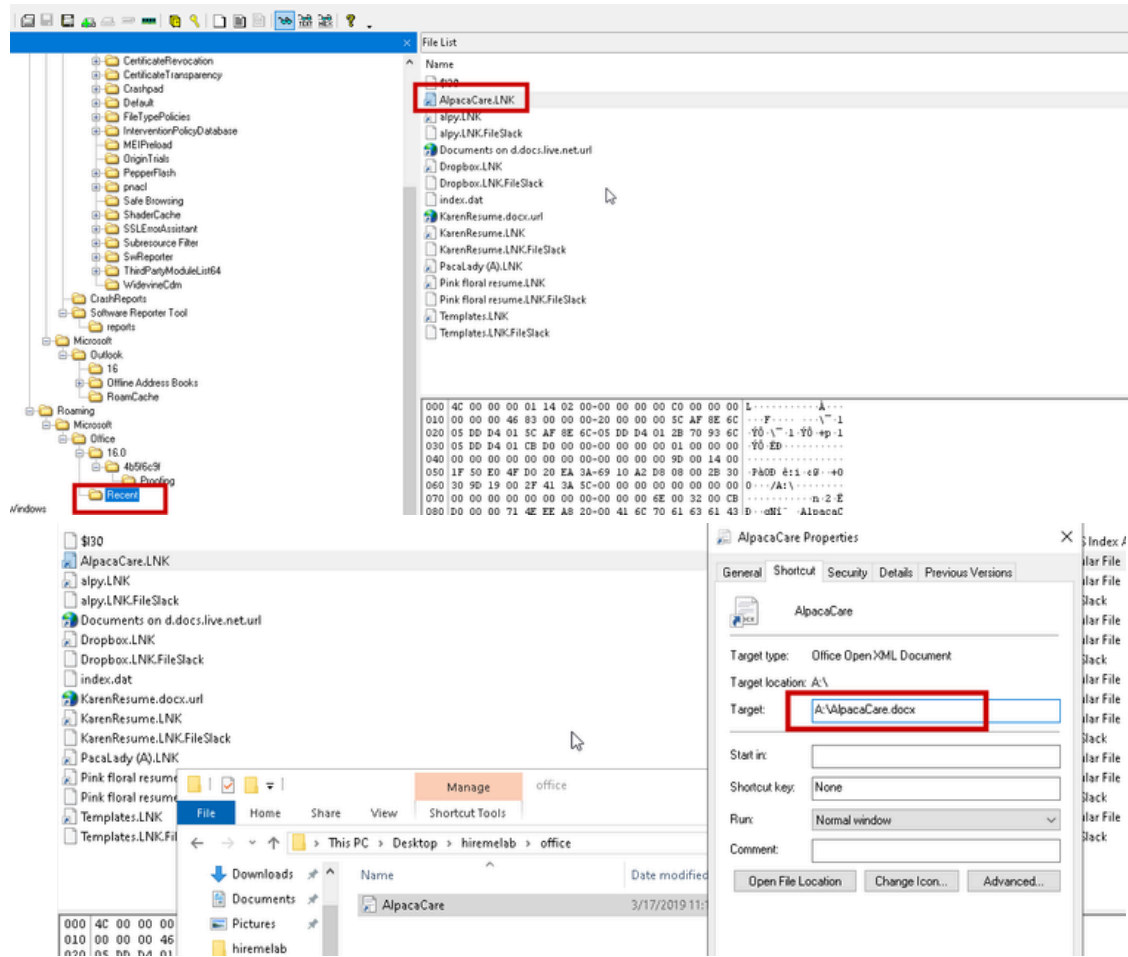
on the second partition looking through it we will find the file and easily just check the date



so the answer here is : 2019-03-17 21:52

Q11-There was a second partition on the drive. What is the letter assigned to it?

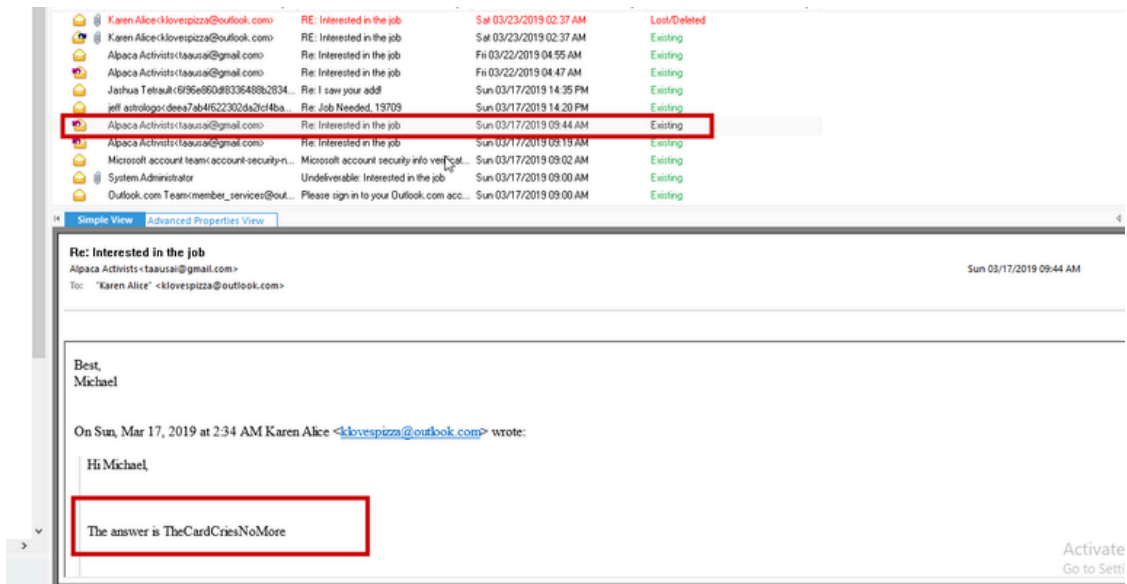
on the first partition looking through it we will find the some LNK files inside the office folder by extracting one of them and viewing the file properties we can find the letter



so the answer here is : A

Q12-What is the answer to the question Company's manager asked Karen?

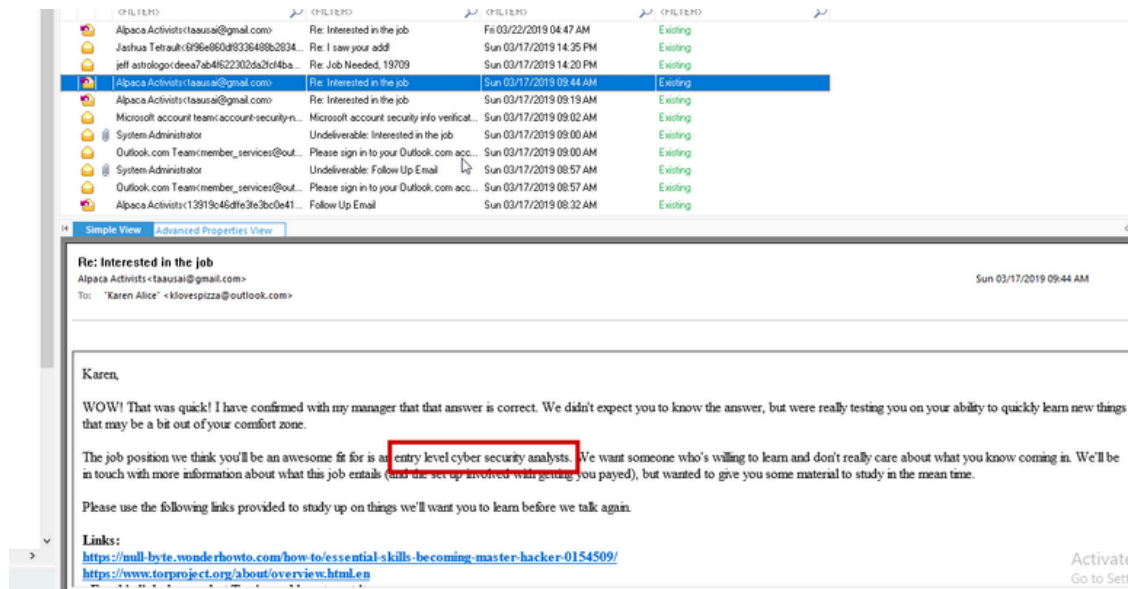
back to OST viewer looking through the emails we will find the answer



so the answer here is : TheCardCriesNoMore

Q13-What is the job position offered to Karen? (3 words, 2 spaces in between)?

looking through the emails we will find the answer



so the answer here is : cyber security analyst

Q14-When was the admin user password last changed?

back on registry explorer passwords are located under the SAM hive exactly under the path
SAM/Domains/Accounts/Users

Values

User accounts

Drag a column header here to group by that column

Valid Us...	User Id	Invalid L...	Total Lo...	Created...	Last Log...	Last Password Change	L...	Expire...	User Name	Full N...	Pass...	Groups	Comm...	User ...	Home...	Inter...	Reset...	Accou...	Home...	Pass...
☒	501	0	0	2019-01...					Guest			Guests	Built-in account for guest access to the computer/domain					☒	☐	☒
☒	503	0	0	2019-01...					DefaultAccount			System Managed Accounts Group	A user account managed by the system.					☒	☐	☒
☒	504	0	0	2019-01...		2019-01-26 19:07:03			WDAGUtilityAccount				A user account managed and used by the system for Windows Defender Application Guard scenarios.					☒	☐	☐
☒	1001	0	32	2019-01...	2019-01...	2019-03-21 19:13:09		2...	Karen			Administrators						☐	☐	☒
☒	1000	0	2	2019-01...	2019-01...	2019-01-26 19:09:41			defaultuser0									☒	☐	☐

so the answer here is : 03/21/2019 19:13:09

Q15-What version of Chrome is installed on the machine?

we need to locate installed apps on the machine and to do that we can look inside the paths either SOFTWARE/Microsoft/Windows/CurrentVersion/Uninstall

or

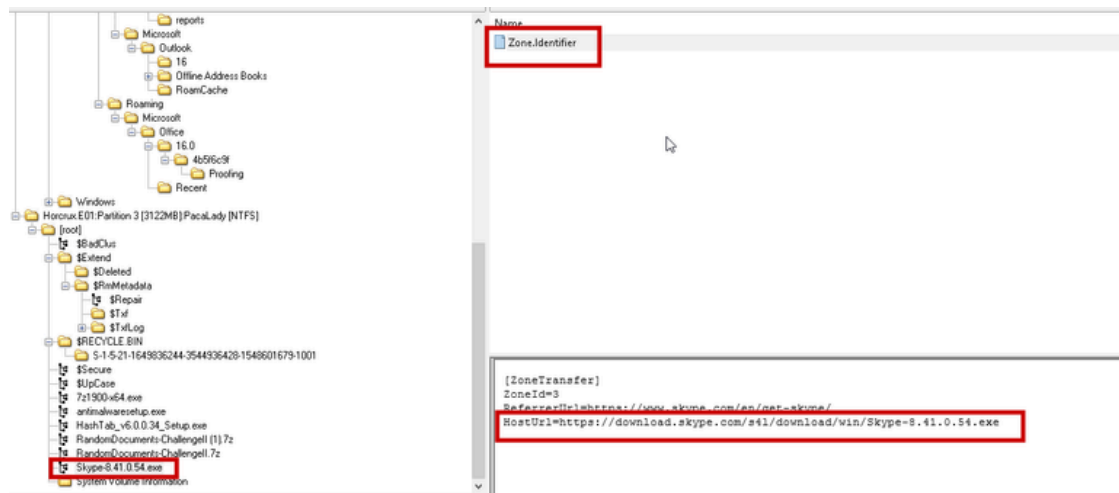
SOFTWARE/WOW6432Node/Microsoft/Windows/CurrentVersion/Uninstall to get the installed apps these are common paths to locate apps. in our case it was under the second path

Path	Name	Type	Data	Timestamp
SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\Google Chrome	DisplayName	RegSz	Google Chrome	
	UninstallString	RegSz	"C:\Program Files (x86)\Google\Chrome\Application\72.0.3626.121\install...	
	InstallLocation	RegSz	C:\Program Files (x86)\Google\Chrome\Application	71-00
	DisplayIcon	RegSz	C:\Program Files (x86)\Google\Chrome\Application\chrome.exe,0	
	NoModify	RegDword	1	
	NoRepair	RegDword	1	
	Publisher	RegSz	Google Inc.	2E-00-31-00
	Version	RegSz	72.0.3626.121	
	DisplayVersion	RegSz	72.0.3626.121	
	InstallDate	RegSz	20190312	04-04
	VersionMajor	RegDword	3626	
	VersionMinor	RegDword	121	

so the answer here is : 72.0.3626.121

Q16-What is the URL used to download Skype?

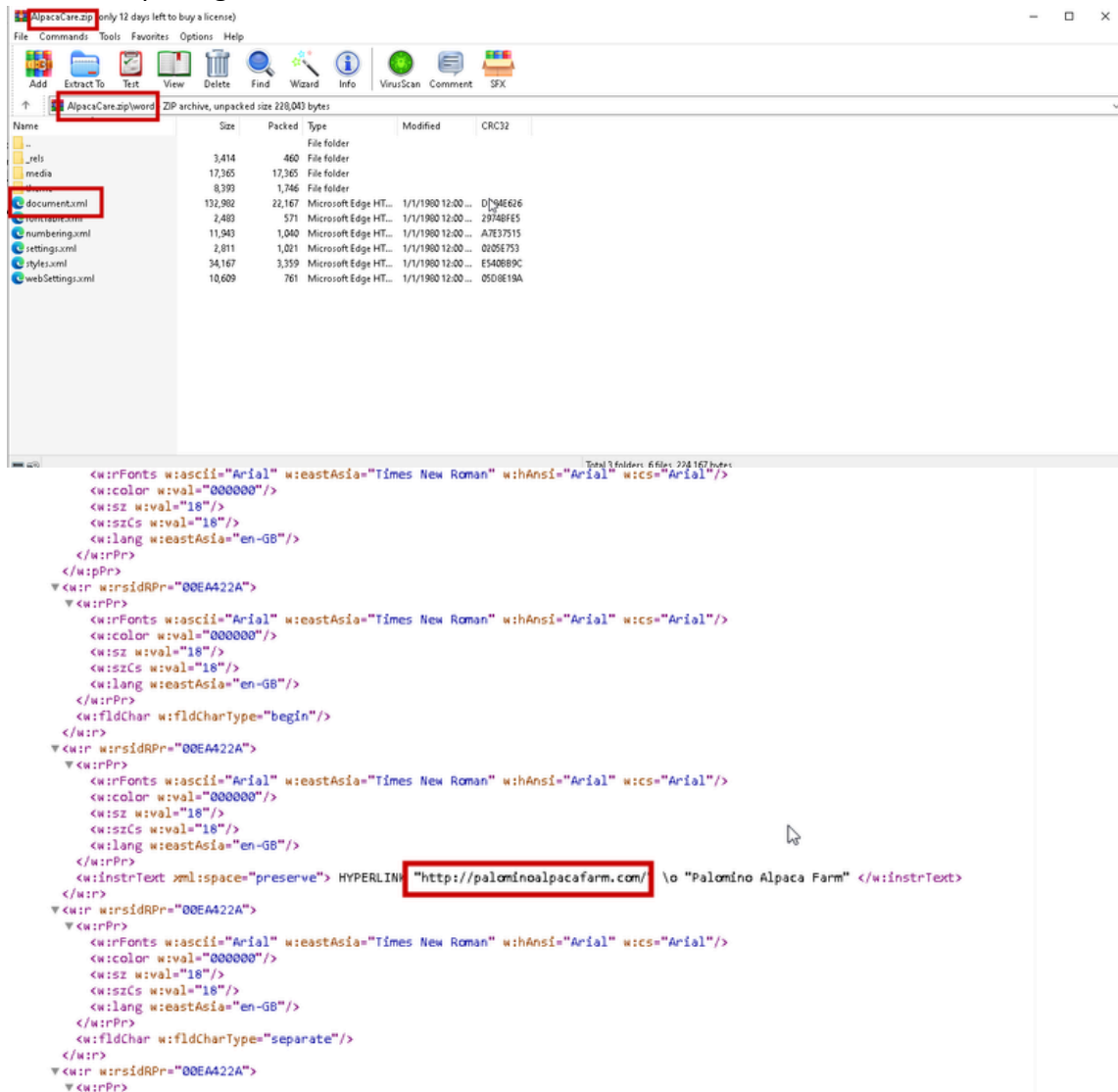
on the second partition if we clicked Skype we can see an alternate data stream called Zone.Identifier opening the file give us the URL



so the answer here is : <https://download.skype.com/s41/download/win/Skype-8.41.0.54.exe>

Q17-What is the domain name of the website Karen browsed on Alpaca care that the file AlpacaCare.docx is based on?

by exporting the file then for better analysis and security we will change the file extension from docx to zip to get all the files without opening the document then looking through it we will find an xml file called document.xml opening it will show us the domain



so the answer here is : palominoalpaca.com

final answers:

- 1- Karen
- 2- 16299
- 3- TOTALLYNOTAHACK
- 4- skype
- 5- 19709
- 6- ms
- 7- 150000
- 8- egypt
- 9- UTC
- 10- 2019-03-17 21:52
- 11- A
- 12- TheCardCriesNoMore
- 13- cyber security analyst
- 14- 03/21/2019 19:13:09
- 15- 72.0.3626.121
- 16- <https://download.skype.com/s4l/download/win/Skype-8.41.0.54.exe>
- 17- palominoalpaca farm.com